WEN, HSING-YU (Winnie)

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EDUCATION

Carnegie Mellon University, Pennsylvania — August 2025 - December 2026
Master of Information Systems Management - Business Intelligence and Data Analytics

Soochow University, Taiwan — September 2019 - June 2023
Bachelor of Business Administration (BBA) in International Business

- GPA: 4.00/4.00 (Class Rank: 1/68; Department Rank: 1/136)

SKILLS & LANGUAGES

- **Programming & AI:** Python, SQL, Java, PyTorch, TensorFlow, scikit-learn, Spark, Git, GitHub
- **Generative AI & NLP:** Large Language Models, RAG, Semantic and Keyword Retrieval, Model Context Protocol, Vector Databases, NLP, Recommendation Systems, LangChain, LangGraph
- **Serving & MLOps:** Model Serving, Evaluation, Monitoring, Drift Detection, Kafka, Docker, Kubernetes, AWS, Databricks

WORK EXPERIENCE

TSMC, Hsinchu City, Taiwan — May 2026 - August 2026
Data AI Engineer Intern

- Engineered an enterprise AI knowledge platform indexing 10,000+ Power BI reports, combining semantic and keyword retrieval for RAG-based natural-language analytics and achieving 90% HitRate@5.
- Automated metadata standardization from REST API outputs through an LLM-powered pipeline, saving an estimated 1,200+ hours of manual documentation.
- Implemented an MCP service mapping natural-language questions to report and dataset IDs for source-aware DAX generation.

Bank SinoPac, Taipei City, Taiwan — June 2024 - June 2025
Data Analyst

- Optimized customer targeting with uplift modeling and A/B testing, increasing marketing ROI by 25% and guiding leadership.
- Delivered 20+ analytics products with Python, T-SQL, and Tableau, improving decision workflows by 47% across digital banking.
- Analyzed a flagship online-to-offline initiative that identified high-potential customers and generated NT$5-10M monthly.

ASUS, Taipei City, Taiwan — May 2023 - November 2023
Data Analyst Intern

- Modeled industry-specific content preferences with logistic regression, increasing conversion by 31% versus the previous campaign.
- Partnered with Deloitte on Salesforce CRM implementation and analytical dashboards, accelerating campaign optimization by 67%.

GoFreight, Taipei City, Taiwan — December 2021 - July 2022
Customer Success Intern

- Forecasted demand with Python and Excel time-series models, improving staffing decisions and reducing personnel costs by 21%.
- Diagnosed system and crawler failures in container-tracking data, increasing accuracy to 98% and reducing manual processing by 20%.

ACADEMIC PROJECTS

Movie Recommendation System — January 2026 - April 2026

- Deployed a Kafka-based recommendation pipeline processing 2,000+ requests in 24 hours and a containerized hybrid inference API with sub-600 ms latency.
- Established data-quality checks, tests, monitoring, drift detection, and offline/online evaluation for recommendation serving.

Email Spam Classification — March 2026

- Vectorized 5,171 imbalanced email samples with CountVectorizer and tuned a k-NN classifier for spam detection.
- Achieved 0.9303 validation weighted F1 and 0.7685 test weighted F1 while evaluating ROC AUC and average precision.

Iterative Processing with Spark — September 2026

- Scaled a Scala/Spark PageRank and recommendation pipeline to 1.0M vertices and 564.8M edges across AWS EMR and Databricks.
- Tuned joins, partitioning, and caching to improve iterative graph-processing efficiency across AWS EMR and Databricks clusters.

NYC Council Bill Clustering — February 2026

- Generated TF-IDF representations for 6,393 bills and compared PCA and UMAP workflows before clustering.
- Benchmarked nine pipelines using k-means, agglomerative clustering, and HDBSCAN to identify interpretable policy themes.